

El espacio topológico producto de dos espacios de Hausdorff es de Hausdorff

Proposición 1. Sean $(X, \mathcal{T}_X)$ e $(Y, \mathcal{T}_Y)$ dos espacios topológicos de Hausdorff

$\implies X \times Y$ con la topología producto es de Hausdorff.

Demostración: Sean $(x_1, y_1), (x_2, y_2) \in X \times Y$ con $(x_1, y_1) \neq (x_2, y_2)$. Como X e Y son de Hausdorff, $\exists U_1, U_2 \in \mathcal{V}(x_1), \mathcal{V}(x_2), V_1, V_2 \in \mathcal{V}(y_1), \mathcal{V}(y_2)$ tales que $U_1 \cap U_2 = \emptyset, V_1 \cap V_2 = \emptyset$. Entonces, $U_1 \times V_1 \in \mathcal{V}((x_1, y_1)) \wedge U_2 \times V_2 \in \mathcal{V}((x_2, y_2))$ y se tiene que

$$(U_1 \times V_1) \cap (U_2 \times V_2) = (U_1 \cap U_2) \times (V_1 \cap V_2) = \emptyset \times \emptyset = \emptyset.$$

Luego $\forall (x_1, y_1), (x_2, y_2) \in X \times Y : (x_1, y_1) \neq (x_2, y_2) : \exists U_1 \times V_1, U_2 \times V_2 \in \mathcal{T}_{X \times Y}$ tales que $(U_1 \times V_1) \cap (U_2 \times V_2) = \emptyset$ y, por tanto, $X \times Y$ es de Hausdorff. ■

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